Q1. $\|(x,y,z)\|_0 = x+y+z$

$\|(1,-1,0)\|_0 = 0$

So we have a non-zero vector v such that, $\|v\|_0 = 0$
Hence, $\|\cdot\|_0$ is not a norm.

$\|(x,y,z)\|_1 = |x|+|y|+|z|$

$\|(x_1,y_1,z_1)+(x_2,y_2,z_2)\|_1 = \|(x_1+x_2, y_1+y_2, z_1+z_2)\|_1$

$= |x_1+x_2| + |y_1+y_2| + |z_1+z_2|$

$\leq |x_1|+|x_2|+|y_1|+|y_2|+|z_1|+|z_2|$

$= \|(x_1,y_1,z_1)\|_1 + \|(x_2,y_2,z_2)\|_1$

$\|c(x,y,z)\|_1 = \|(cx,cy,cz)\|_1 = |cx|+|cy|+|cz|$

$= |c|\left(|x|+|y|+|z|\right)$

$= |c|\,\|(x,y,z)\|_1$

$\|(x,y,z)\|_1 = 0$

$\Rightarrow |x|+|y|+|z| = 0 \Rightarrow x = 0 = y = z \Rightarrow (x,y,z) = (0,0,0)$

and $\|(0,0,0)\|_1 = 0$

Hence, $\|\cdot\|_1$ is a norm.

$$\|(x, y, z)\|_2 = \sqrt{x^2 + y^2 + z^2}$$

$$\|(x_1, y_1, z_1) + (x_2, y_2, z_2)\|_2$$

$$= \|(x_1 + x_2, y_1 + y_2, z_1 + z_2)\|_2$$

$$= \sqrt{(x_1 + x_2)^2 + (y_1 + y_2)^2 + (z_1 + z_2)^2}$$

$$\leq \sqrt{x_1^2 + y_1^2 + z_1^2} + \sqrt{x_2^2 + y_2^2 + z_2^2}$$

$$= \|(x_1, y_1, z_1)\|_2 + \|(x_2, y_2, z_2)\|_2$$

$$\|c(x, y, z)\|_2 = \|(cx, cy, cz)\|_2 = \sqrt{c^2 x^2 + c^2 y^2 + c^2 z^2}$$

$$= |c| \sqrt{x^2 + y^2 + z^2}$$

$$= |c| \, \|(x, y, z)\|_2$$

$$\|(x, y, z)\|_2 = 0$$

$$\Rightarrow \sqrt{x^2 + y^2 + z^2} = 0$$

$$\Rightarrow x^2 + y^2 + z^2 = 0 \Rightarrow x = y = z = 0 \Rightarrow (x, y, z) = (0, 0, 0)$$

$$\|(0, 0, 0)\|_2 = 0 .$$

Hence, $\|\cdot\|_2$ is a norm.

$$\|(x,y,z)\|_3 = \max\{|x|,|y|,|z|\}$$

$$\|(x_1,y_1,z_1)+(x_2,y_2,z_2)\|_3$$

$$= \|(x_1+x_2,\ y_1+y_2,\ z_1+z_2)\|_3$$

$$= \max\{|x_1+x_2|,|y_1+y_2|,|z_1+z_2|\}$$

$$\leq \max\{|x_1|,|y_1|,|z_1|\} + \max\{|x_2|,|y_2|,|z_2|\}$$

$$= \|(x_1,y_1,z_1)\|_3 + \|(x_2,y_2,z_2)\|_3$$

$$\|c(x,y,z)\|_3 = \|(cx,cy,cz)\|$$

$$= \max\{|cx|,|cy|,|cz|\}$$

$$= |c|\max\{|x|,|y|,|z|\}$$

$$= |c|\,\|(x,y,z)\|_3$$

$$\|(x,y,z)\|_3 = 0$$

$$\Rightarrow \max\{|x|,|y|,|z|\} = 0$$

$$\Rightarrow |x| = |y| = |z| = 0 \Rightarrow x = y = z = 0 \Rightarrow (x,y,z) = (0,0,0)$$

$\|(0,0,0)\|_3 = 0$. Hence, $\|\cdot\|_3$ is a norm.

$V_2 = M_{2\times 2}(\mathbb{R})$ $\qquad A \in V_2 \qquad A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$

$\|A\|_4 = \max\{|a_{11}| + |a_{12}|, |a_{21}| + |a_{22}|\}$

$\|A\|_4 = 0 \Rightarrow \max\{|a_{11}| + |a_{12}|, |a_{21}| + |a_{22}|\} = 0$

$\Rightarrow |a_{11}| + |a_{12}| = 0, \ |a_{21}| + |a_{22}| = 0$

$\Rightarrow a_{11} = a_{12} = a_{21} = a_{22} = 0$

$A = 0$

conversely,

If $A = 0$, then $\|A\|_4 = 0$.

$\|cA\|_4 = \max\{|ca_{11}| + |ca_{12}|, |ca_{21}| + c|a_{22}|\}$

$= \max\{|c|(|a_{11}| + |a_{12}|), |c|(|a_{21}| + |a_{22}|)\}$

$= |c| \max\{|a_{11}| + |a_{12}|, |a_{21}| + |a_{22}|\}$

$= |c| \, \|A\|_4$.

$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \qquad B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$

$A + B = \begin{pmatrix} a_{11} + b_{11} & a_{12} + b_{12} \\ a_{21} + b_{21} & a_{22} + b_{22} \end{pmatrix}$

$$\|A+B\|_4 = \max\{|a_{11}+b_{11}|+|a_{12}+b_{12}|,\ |a_{21}+b_{21}| + |a_{22}+b_{22}|\}$$

$$\leq \max\{|a_{11}|+|b_{11}|+|a_{12}|+|b_{12}|,\ |a_{21}|+|b_{21}|+|a_{22}|+|b_{22}|\}$$

$$\leq \max\{|a_{11}|+|a_{12}|,\ |a_{21}|+|a_{22}|\} + \max\{|b_{11}|+|b_{12}|,\ |b_{21}|+|b_{22}|\}$$

$$\leq \|A\|_4 + \|B\|_4 .$$

$\|\cdot\|_4$ is a norm.

$\|\cdot\|_5$ Check similarly.

Q2) $T : \mathbb{R}^3 \to \mathbb{R}$

$T(u) = \langle u, v \rangle$ where $v \in \mathbb{R}^3$.

Suppose T is one to one, then $\text{nullity}(T) = 0$

In this case,

$\text{nullity}(T) + \text{rank}(T) = \dim(\mathbb{R}^3) = 3$

$\Rightarrow 0 + \text{rank}(T) = 3$

$\Rightarrow \text{rank}(T) = 3$

$\text{rank}(T)$ can not be 3 as $\dim(\text{codomain}(T)) = 1$.

Hence we have a contradiction.

So, T is not one to one.

If $v \neq 0 \in \mathbb{R}^3$, then

$T(v) = \langle v, v \rangle \neq 0 \in \mathbb{R}$

$\Rightarrow$ image of T is non zero.

$\Rightarrow \text{Rank}(T) \geq 1$

Since, $\dim(\mathbb{R}) = 1 \Rightarrow \text{image}(T) = \mathbb{R}$.

So, T is onto.

T is not one to one $\Rightarrow$ T is not an isomorphism.

Q3. a) $V = \mathbb{R}^2$, $\langle (x_1, x_2), (y_1, y_2) \rangle = \sum_{i,j=1}^{2} x_i y_j$

$$= x_1 y_1 + x_2 y_2 + x_2 y_1 + x_1 y_2$$

$(1, -1) \neq (0,0) \in \mathbb{R}^2$, but

$\langle (1, -1), (1, -1) \rangle = 0$

Hence, this cannot be an inner product.

b) $V = M_{2\times 2}(\mathbb{R})$

$$\langle x+y, z \rangle = \langle x+y, z \rangle_1 + \langle x+y, z \rangle_2$$

$$= \langle x, z \rangle_1 + \langle y, z \rangle_1 + \langle x, z \rangle_2 + \langle y, z \rangle_2$$

$$= \langle x, z \rangle_1 + \langle x, z \rangle_2 + \langle y, z \rangle_1 + \langle y, z \rangle_2$$

$$= \langle x, z \rangle + \langle y, z \rangle$$

$$\langle cx, y\rangle = \langle cx, y\rangle_1 + \langle cx, y\rangle_2$$

$$= c\langle x, y\rangle_1 + c\langle x, y\rangle_2$$

$$= c\left(\langle x, y\rangle_1 + \langle x, y\rangle_2\right)$$

$$= c\langle x, y\rangle$$

Similarly we can check the other conditions for an inner product space.

c) Take $\langle\, ,\rangle_1 = \langle\, ,\rangle_2$ then

$$\langle x, y\rangle = \langle x, y\rangle_1 - \langle x, y\rangle_2 = 0$$

for any $x, y \in V$.

d) $V = \mathbb{R}^2$ $\quad \langle (x_1, x_2), (y_1, y_2)\rangle = x_1 y_2 - x_2 y_1$

$\langle (1,1), (1,1)\rangle = 0$, but $(1,1) \neq (0,0)$

This cannot be an inner product on V.

Q.4) $T(x,y,z) = (x+2y,\ x-y+cz,\ 2x+y+dz)$

$T(1,0,0) = (1,1,2)$

$T(0,1,0) = (2,-1,1)$

$T(0,0,1) = (0,c,d)$

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 1 & -1 & c \\ 2 & 1 & d \end{pmatrix}$$

$$B = \begin{pmatrix} 0 & 3 & 2 \\ -1 & -3 & -3 \\ 2 & 3 & 4 \end{pmatrix}$$

$$B = \begin{pmatrix} 0 & 3 & 2 \\ -1 & -3 & -3 \\ 2 & 3 & 4 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} -1 & -3 & -3 \\ 0 & 3 & 2 \\ 2 & 3 & 4 \end{pmatrix}$$

$$\xrightarrow{-R_1} \begin{pmatrix} 1 & 3 & 3 \\ 0 & 3 & 2 \\ 2 & 3 & 4 \end{pmatrix}$$

$$\xrightarrow{R_3 - 2R_1} \begin{pmatrix} 1 & 3 & 3 \\ 0 & 3 & 2 \\ 0 & -3 & -2 \end{pmatrix}$$

$$\xrightarrow{R_3 + R_2} \begin{pmatrix} 1 & 3 & 3 \\ 0 & 3 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow{R_2/3} \begin{pmatrix} 1 & 3 & 3 \\ 0 & 1 & 2/3 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow{R_1 - 3R_2} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 2/3 \\ 0 & 0 & 0 \end{pmatrix}$$

Rank of B is $2 \Rightarrow$ Rank of A is 2 (as A and B is similar)

$\Rightarrow$ Rank of T is 2.

nullity of A is $1 \Rightarrow$ nullity of T is 1.

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 1 & -1 & c \\ 2 & 1 & d \end{pmatrix} \xrightarrow[R_3 - 2R_1]{R_2 - R_1} \begin{pmatrix} 1 & 2 & 0 \\ 0 & -3 & c \\ 0 & -3 & d \end{pmatrix}$$

$$\xrightarrow{R_3 - R_2} \begin{pmatrix} 1 & 2 & 0 \\ 0 & -3 & c \\ 0 & 0 & d-c \end{pmatrix} \xrightarrow{R_2/-3} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & -c/3 \\ 0 & 0 & d-c \end{pmatrix}$$

If $c \neq d$, then $d - c \neq 0$, i.e. $\text{rank}(A) = 3$

Then A cannot be similar to B.

If A is similar to B then $\text{rank}(A) = 2$

In that case, $d - c = 0$

i.e. $d = c$.

and $c = 1$ is given.

The row echelon form of A is,

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & -1/3 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 - 2R_1} \begin{pmatrix} 1 & 0 & 2/3 \\ 0 & 1 & -1/3 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\left.\begin{aligned} x + \tfrac{2}{3} z &= 0 \\ y - \tfrac{1}{3} z &= 0 \end{aligned}\right\} \Rightarrow \begin{aligned} x &= -\tfrac{2}{3} z \\ y &= \tfrac{1}{3} z \end{aligned}$$

$$\ker(T) = \left\{ \left(-\tfrac{2}{3} z, \tfrac{1}{3} z, z\right) \middle| z \in \mathbb{R} \right\}$$

$$\text{Basis of } \ker(T) = \left\{ \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} \right\}.$$

Q.5) $v=(1,2)$ $\quad \|v\| = \sqrt{v \cdot v}$

$$= \sqrt{1+4}$$

$$= \sqrt{5}$$

<u>option 1</u>

Let $u=(2,1)$ then, $\|u\| = \sqrt{u \cdot u}$

$$= \sqrt{1+4} = \sqrt{5}$$

So, $\|u\| = \|v\|$.

Hence, there exists a vector u in $\mathbb{R}^2$, such that $\|u\| = \|v\|$

<u>option 2:</u>

Consider the set $\{(x,y) \mid x^2+y^2=5\}$

Hence, there are infinitely many vectors in $\mathbb{R}^2$ such that norms of the vectors are equal to norm of v.

<u>Option 3:</u> Consider $u=(-2,1)$

$$\langle u, v\rangle = (-2,1)\cdot(1,2) = -2+2=0$$

Option 4: Consider the set

$$\{ c(-2,1) \mid c \in \mathbb{R} \}$$

Hence, there are infinitely many vectors in $\mathbb{R}^2$ such that angle between the vector and v is equal to $90°$.

Q6) Take a look at "Reflect with us" of week 7 in the portal.

Q7) $L_1 = \{ (x,y,z) \mid x + y + 2z = 6 \}$

$L_2 = \{ (x,y,z) \mid 2x + y + 2z = 9 \}$

W_1 is the corresponding vector subspace of L_1

$W_1 = \{ (x,y,z) \mid x + y + 2z = 0 \}$

W_2 is the corresponding vector subspace of L_2

$W_2 = \{ (x,y,z) \mid 2x + y + 2z = 0 \}$

$$W_1 \cap W_2 = \{(x,y,z) \mid x+y+2z=0 \text{ and } 2x+y+2z=0\}$$

$$x+y+2z=0$$

$$2x+y+2z=0$$

$$\begin{pmatrix} 1 & 1 & 2 \\ 2 & 1 & 2 \end{pmatrix} \xrightarrow{R_2-2R_1} \begin{pmatrix} 1 & 1 & 2 \\ 0 & -1 & -2 \end{pmatrix}$$

$$\Big\downarrow -R_2$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \end{pmatrix} \xleftarrow{R_1-R_2} \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 2 \end{pmatrix}$$

$$x=0$$

$$y+2z=0$$

Hence, $W_1 \cap W_2 = \{(x,y,z) \mid x=0 \ , \ y+2z=0\}$

Q8) $u=(2,1,-3)$

$v=(5,8,4)$

$$u \cdot v = \|u\|\|v\| \cos\theta$$

$$\Rightarrow (2,1,-3)\cdot(5,8,4) = \|u\|\|v\|\cos\theta$$

$\Rightarrow \quad 10 + 8 - 12 = \|u\| \, \|v\| \cos\theta$

$\Rightarrow \quad \|u\| \, \|v\| \cos\theta = 6$

8,9) $\quad v_1 = (1, 0, -1) \qquad v_2 = (a, b, c)$

$\|v_2\| = 3$

$\langle v_2, (1, -1, 1) \rangle = -3$

$\Rightarrow \quad a - b + c = -3$

$\langle v_2, (-1, 2, 1) \rangle = 1$

$\Rightarrow \quad -a + 2b + c = 1$

$v_1 \cdot v_2 = \|v_1\| \, \|v_2\| \cos\theta$

$\Rightarrow \quad a - c = \sqrt{2} \cdot 3 \cdot \frac{1}{\sqrt{2}}$

$\Rightarrow \quad a - c = 3$

$$a - b + c = -3$$
$$-a + 2b + c = 1$$
$$a - c = 3$$

Adding both the sides of all the three equations we have,

$$a + b + c = 1$$

Q10) Rank of A is 3

A and B are equivalent.

So, Rank of B = Rank of A = 3

B and C are similar.

So, $\det(B) = \det(C)$

B is a 4X4 matrix with rank 3,

hence $\det(B) = 0$

Hence, $\det(C) = 0$